Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Find the area bounded by the curve $y = 2x - x^2$ & the straight line $y = -x$.

वक्र $y = 2x - x^2$ तथा सरल रेखा $y = -x$ द्वारा परिबद्ध क्षेत्रफल होगा।

Ans. $\frac{9}{2}$

Sol. $y = 2x - x^2$

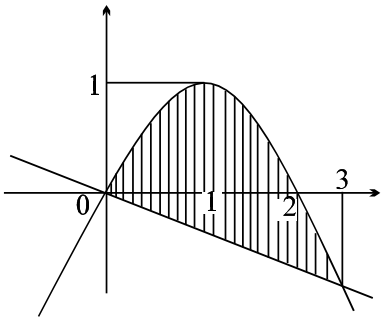
$$\frac{dy}{dx} = 2 - 2x = 0 \Rightarrow x = 1$$

solving $y = 2x - x^2$ and $y = -x$, we have

$$-x = 2x - x^2 \Rightarrow x^2 - 3x = 0 \Rightarrow x = 0 \text{ or } 3$$

$$A = \int_0^3 \left[\left(2x - x^2\right) - (-x)\right] dx = \int_0^3 \left(3x - x^2\right) dx$$

$$= \frac{3x^2}{2} - \frac{x^3}{3}\Big]_0^3 = \frac{27}{2} - 9 = \frac{9}{2} \text{ sq. unit}$$



2. Find the area enclosed by the parabola $y = 1 + x^2$ and a normal drawn to it with gradient -1.

परवलय $y = 1 + x^2$ तथा इसके अभिलम्ब जिसका ढाल -1 है द्वारा परिबद्ध क्षेत्रफल होगा।

Ans. $\frac{4}{3}$

Sol. $y = 1 + x^2 \quad \Rightarrow \quad \frac{dy}{dx} = 2x$ [Ans. $A = \frac{4}{3}$ sq. units]

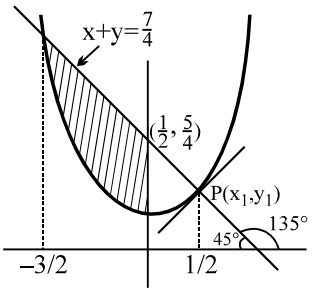
Hence $m = -\frac{1}{2x_1}$ (m = slope of normal at x_1, y_1)

$$\therefore \quad x_1 = \frac{-1}{2m} = \frac{1}{2} \quad ; \quad y_1 = 1 + \frac{1}{4m^2} = \frac{5}{4}$$

equation of normal $\quad y - \left(1 + \frac{1}{4m^2}\right) = m\left(x + \frac{1}{2m}\right)$

put $m = -1$, we get $y - \frac{5}{4} = -\left(x - \frac{1}{2}\right)$

$$x + y = \frac{5}{4} + \frac{1}{2} = \frac{7}{4} \qquad \ldots.(1)$$



$$A = \int_{x_1}^{x_2}\left(\frac{7}{4} - x\right) - (1 + x^2)\,dx = \int_{x_1}^{x_2}\left(\frac{3}{4} - x - x^2\right)dx$$

now $1 + x^2 = -x + \frac{7}{4}$

$4x^2 + 4x - 3 = 0 \Rightarrow x = -\frac{3}{2}$ and $x = \frac{1}{2}$

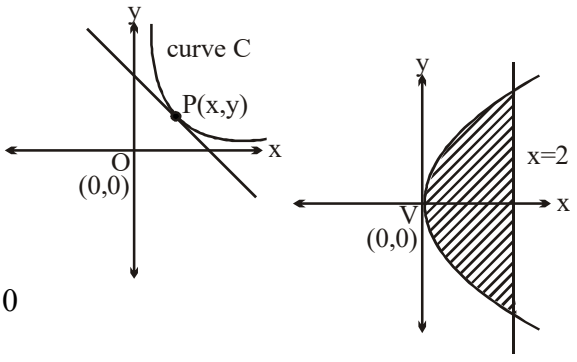
$\therefore \quad A = \int_{-3/2}^{1/2}\left(\frac{3}{4} - x - x^2\right)dx$ simplifying $A = \frac{4}{3}$ sq. units

3. Let C be a curve passing through M (2, 2) such that the slope of the tangent at any point to the curve is reciprocal of the ordinate of the point. If the area bounded by curve C and line $x = 2$ is expressed as a rational $\frac{p}{q}$ (where p and q are in their lowest form), then find $(p + q)$.

वक्र C बिन्दु M (2, 2) से गुजरता है जिसके किसी बिन्दु पर स्पर्श रेखा का ढाल उस बिन्दु की कोटि का व्युत्क्रम है। यदि वक्र C तथा रेखा $x = 2$ द्वारा परिबद्ध क्षेत्रफल $\frac{p}{q}$ है। (जहाँ p तथा q सह अभाज्य है।), तो $(p + q)$ का मान ज्ञात कीजिए।

Ans. 19

Sol. Let P (x, y) be any point on the curve C.



Now, $\frac{dy}{dx} = \frac{1}{y}$

$\Rightarrow ydy = dx \Rightarrow \frac{y^2}{2} = x + k$

Since the curve passes through M (2,2), so $k = 0$

$\Rightarrow y^2 = 2x$

Hence required area $= 2\int_0^2 \sqrt{2x}\,dx = 2\sqrt{2} \times \frac{2}{3}\left(x^{3/2}\right)_0^2 = \frac{4}{3}\sqrt{2} \times 2\sqrt{2} = \frac{16}{3}$ (square unit)

$\Rightarrow \quad p + q = 19$

4. Let l be the line created by rotating the tangent line to the parabola $y = x^2$ at the point A(1, 1), about A by an angle of $(-\pi/4)$. Let B be the other intersection of the line l with $y = x^2$. If the area enclosed by the line l and the parabola is $\frac{a_1}{a_2}$ where a_1 and a_2 are coprime, find $(a_1 + a_2)$.

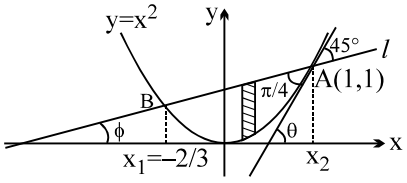
परवलय $y = x^2$ के बिन्दु A(1, 1) पर खीचीं स्पर्श रेखा को बिन्दु A के सापेक्ष $(-\pi/4)$ कोण घुमा देने पर नई रेखा l पुनः परवलय $y = x^2$ को बिन्दु B पर काटती है। यदि रेखा l तथा परवलय द्वारा परिबद्ध क्षेत्रफल $\frac{a_1}{a_2}$ हो (a_1 एवं a_2 सापेक्षतः अभाज्य है) तो $(a_1 + a_2)$ ज्ञात कीजिए।

Ans. 287

Sol. $y = x^2$

$$\left.\frac{dy}{dx}\right]_{A(1,1)} = 2x = 2; \quad \text{hence, } \tan\theta = 2 \qquad ....(1)$$

Let the line l makes an angle ϕ with the x-axis



then, $\phi = \theta - \frac{\pi}{4}; \quad m_l = \tan\phi = \frac{\tan\theta - 1}{1 + \tan\theta} = \frac{1}{3}$

$\therefore$ equation of the line l is,

$$y - 1 = \frac{1}{3}(x - 1) \quad \Rightarrow \quad 3y - 3 = x - 1 \Rightarrow \quad 3y = x + 2$$

solving line and parabola $y = x^2$,

$$3x^2 = x + 2 \Rightarrow 3x^2 - x - 2 = 0$$

$$x_1x_2 = -\frac{2}{3}; \quad \text{but } x_2 = 1; \quad \therefore \quad x_1 = -\frac{2}{3}$$

$$\therefore \quad A = \int_{-2/3}^{1}\left(\frac{x+2}{3} - x^2\right)dx = \frac{1}{3}\int_{-2/3}^{1}\left(x + 2 - 3x^2\right)dx = \frac{1}{3}\left[\frac{x^2}{2} + 2x - x^3\right]_{-2/3}^{1}$$

$$= \frac{1}{3}\left[\left(\frac{1}{2} + 2 - 1\right) - \left(\frac{2}{9} - \frac{4}{3} + \frac{8}{27}\right)\right] = \frac{1}{3}\left[\frac{3}{2} - \left(\frac{6 - 36 + 8}{27}\right)\right] = \frac{1}{3}\left[\frac{3}{2} + \frac{22}{27}\right]$$

$$= \frac{1}{3}\left[\frac{81 + 44}{54}\right] = \frac{125}{162} = \frac{a_1}{a_2} \quad \Rightarrow \quad a_1 + a_2 = 287$$

5. Let $f(x) = x - x^2$ and $g(x) = ax, a < 0$. If the area bounded by $f(x)$ and $g(x)$ is equal to the area bounded by the curves $x = 3y - y^2$ and $x + y = 3$, then find the value of $|[a]|$

[Note : [k] denotes the greatest integer less than or equal to k.]

माना $f(x) = x - x^2$ एवं $g(x) = ax$ है। यदि $f(x)$ व $g(x)$ द्वारा परिबद्ध क्षेत्रफल वक्रों $x = 3y - y^2$ एवं $x + y = 3$ द्वारा परिबद्ध क्षेत्रफल के बराबर है, तो $|[a]|$ का मान ज्ञात कीजिए।

[नोट : $[k]$ महत्तम पूर्णांक को दर्शाता है जो 'k' के बराबर व 'k' से छोटा है]

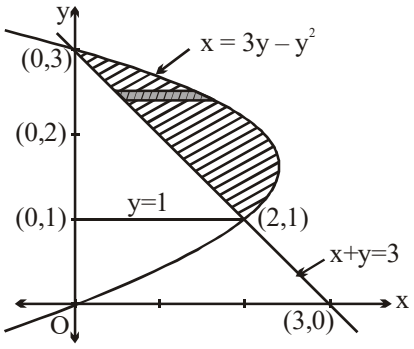
Ans. 1

Sol. Area between $x = 3y - y^2$ and $x + y = 3$

$$A = \int_1^3 \left(3y - y^2\right) - (3 - y)\, dy$$

$$= 4 \cdot \frac{y^2}{2} - \frac{y^3}{3} - 3y \Big]_1^3 = (2 \cdot 9 - 9 - 9) - \left(2 - \frac{1}{3} - 3\right)$$

$$A = \frac{4}{3}$$



Now area bounded by $f(x)$ above the x-axis is $= \int_0^1 \left(x - x^2\right) dx = \frac{1}{6}$, hence $a < 0$ (think)

Given, $\int_0^{1-a} \left((x - x^2) - (a\,x)\right) dx = \frac{4}{3}$

Solving $\frac{1}{6}(1 - a)^3 = \frac{4}{3}$

$(1 - a)^3 = 8$

$1 - a = 2 \Rightarrow a = -1$

$\Rightarrow \quad |[a]| = 1$

Note :

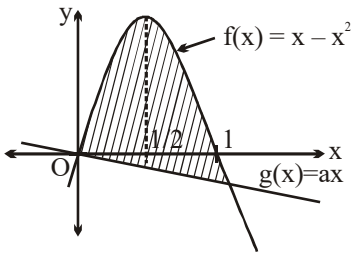
$x - x^2 = ax$

$x^2 + (a - 1)x = 0$

$x_1 + x_2 = 1 - a$

but $x_1 = 0$

$\Rightarrow x_2 = 1 - a$

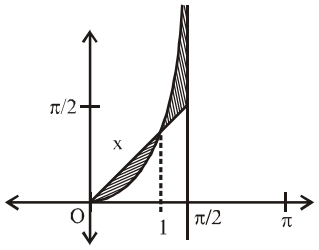


6. If the area bounded by the curves $f(x) = \left(\cos^{-1}|\cos x|\right)^2$, $g(x) = \left(\cos^{-1}|\cos x|\right)$ and $|x| = \frac{\pi}{2}$ is $a\pi^3 + b\pi^2 + c$, then find the minimum value of $(|a| + |b| + |c|)$.

यदि वक्रों $f(x) = \left(\cos^{-1}|\cos x|\right)^2$, $g(x) = \left(\cos^{-1}|\cos x|\right)$ एवं $|x| = \frac{\pi}{2}$ द्वारा परिबद्ध क्षेत्रफल $a\pi^3 + b\pi^2 + c$ है, तो $(|a| + |b| + |c|)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 1

Sol. We have $f(x) = \begin{bmatrix} x^2 ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x)^2 ; & x \in \left(\frac{\pi}{2}, \pi\right) \end{bmatrix}$; $g(x) = \begin{bmatrix} x ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x); & x \in \left(\frac{\pi}{2}, \pi\right) \end{bmatrix}$



Both f and g are periodic with period = π and symmetric about x = π/2

$\therefore$ Required area $= 2\left(\int_0^1 (x - x^2)\,dx + \int_1^{\pi/2} (x^2 - x)\,dx\right)$

$$= 2\left(\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{\pi^3}{24} - \frac{1}{3}\right) - \left(\frac{\pi^2}{8} - \frac{1}{2}\right)\right)$$

$$= 2\left(\frac{\pi^3}{24} - \frac{\pi^2}{8} + \frac{1}{3}\right) = \frac{\pi^3}{12} - \frac{\pi^2}{4} + \frac{2}{3} \equiv a\pi^3 + b\pi^2 + c \text{ (Given)}$$

So, $a = \frac{1}{12}$, $b = \frac{-1}{4}$, $c = \frac{2}{3}$

Hence, $(|a| + |b| + |c|) = \left|\frac{1}{12}\right| + \left|\frac{-1}{4}\right| + \left|\frac{2}{3}\right| = \frac{1}{12} + \frac{1}{4} + \frac{2}{3} = \frac{1+3+8}{12} = \frac{12}{12} = 1$

7. A polynomial function $f(x)$ satisfies the condition $f(x+1) = f(x) + 2x + 1$. Find $f(x)$ if $f(0) = 1$. Find also the equations of the pair of tangents from the origin on the curve $y = f(x)$ and compute the area enclosed by the curve and the pair of tangents.

एक बहुपदीय फलन $f(x)$, $f(x+1) = f(x) + 2x + 1$ का सन्तुष्ट करता है। यदि $f(0) = 1$ तब $f(x)$ ज्ञात कीजिए। साथ ही मूल बिन्दु से वक्र $y = f(x)$ पर स्पर्श रेखा युग्म के समीकरणों का ज्ञात कीजिए तथा वक्र $y = f(x)$ और स्पर्श रेखाओं से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

Ans. $f(x) = x^2 + 1$; $y = \pm 2x$; $A = \frac{2}{3}$ sq. units

Sol. $f(x+1) = f(x) + 2x + 1$

$\Rightarrow f''(x+1) = f''(x) \quad \forall x \in R$

Let $f''(x) = a \quad \Rightarrow \quad f'(x) = ax + b$

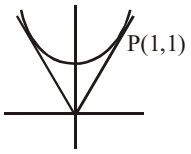
$\Rightarrow f(x) = \frac{ax^2}{2} + bx + c$

$\Rightarrow c = 1 \quad [\because f(0) = 1]$

Now $f(x+1) - f(x) = 2x + 1$

$\Rightarrow \left[\frac{a}{2}(x+1)^2 + b(x+1) + c\right] - \left[\frac{ax^2}{2} + bx + c\right] = 2x + 1$

$\Rightarrow ax + \frac{a}{2} + b = 2x + 1$



on comparing we get $a = 2$,

or $\frac{a}{2} + b = 1 \Rightarrow b = 0$

$\therefore f(x) = x^2 + 1$ ... (i)

Now let equation of tangent be $y = mx$ ... (ii)

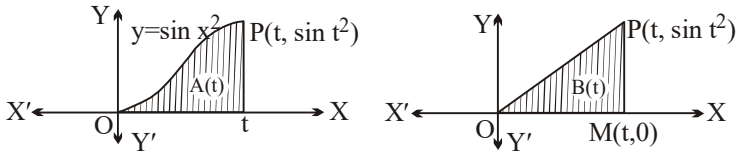
from (i) and (ii)

$x^2 - mx + 1 = 0 \Rightarrow m = \pm 2$

$\therefore$ tangent are $y = 2x$ or $y = -2x$

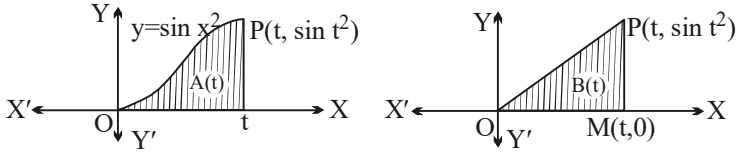
$$A = 2\int_0^1 (x^2 + 1 - 2x)dx = \frac{2}{3}$$

8. The figure shows two regions in the first quadrant.



A(t) is the area under the curve $y = \sin x^2$ from 0 to t and B(t) is the area of the triangle with vertices O, P and M(t, 0). Find $\lim_{t\to 0} \frac{A(t)}{B(t)}$.

चित्र प्रथम चतुर्थांश में दो क्षेत्रो को दर्शाते है।



वक्र $y = \sin x^2$ का $x = 0$ से t में क्षेत्रफल A(t) और B(t) उस त्रिभुज का क्षेत्रफल जिसके शीर्ष O, P और M(t, 0) है तब $\lim_{t\to 0} \frac{A(t)}{B(t)}$ का मान ज्ञात कीजिए।

Ans. 2/3

Sol. $A(t) = \int_0^t \sin x^2 dx$ & $B(t) = \frac{1}{2} \times t \sin t^2$

$$\lim_{t\to 0}\frac{A(t)}{B(t)}=\lim_{t\to 0}\frac{\int_0^t \sin x^2 dx}{\frac{1}{2}t\cdot\sin t^2}$$

$$=\lim_{t\to 0}\left[\frac{2\int_0^t \sin x^2 dx}{t^3}\right]$$

DLH

$$=\lim_{t\to 0}\frac{2\sin t^2}{3t^2}$$

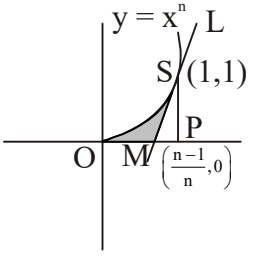
$$=\frac{2}{3}$$

9. Consider the curve $y = x^n$ where $n > 1$ in the 1st quadrant. If the area bounded by the curve, the x-axis and the tangent line to the graph of $y = x^n$ at the point (1, 1) is maximum then find the value of n.

माना प्रथम चतुर्थांशों में वक्र $y = x^n$ जहाँ $n > 1$ है। यदि वक्र $y = x^n$, x-अक्ष और बिन्दु (1, 1) पर स्पर्श रेखा से परिबद्ध क्षेत्र का क्षेत्रफल अधिकतम है तब n का मान ज्ञात कीजिए।

Ans. $\sqrt{2}+1$

Sol. equation of line L tangent at (1,1)



$y = x^n$

$$\frac{dy}{dx}=nx^{n-1}$$

$$m_T=n$$

$$L: y=nx-n+1 \Rightarrow \text{x-intercept}\left(\frac{n-1}{n},0\right)$$

shaded area

$$A=\int_0^1 x^n dx-\frac{1}{2}\times\left(1-\frac{n-1}{n}\right)\times 1$$

$$A = \frac{1}{n+1} - \frac{1}{2}\left(\frac{1}{n}\right)$$

$$A = \frac{1}{n+1} - \frac{1}{2n}$$

$$A = \frac{n-1}{2n(n+1)}$$

$$\frac{dA}{dn} = \frac{2n(n+1) - 2(n-1)n - 2(n-1)(n+1)}{4n^2(n+1)^2} = 0$$

$= n^2 + n - n^2 + n - n^2 + 1 = 0$

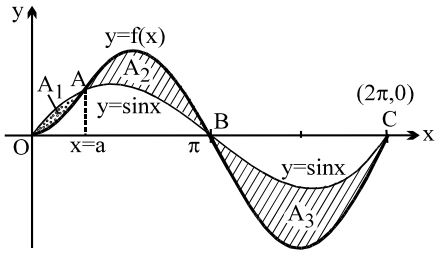
$-n^2 + 2n + 1 = 0$

$n^2 - 2n - 1 = 0$

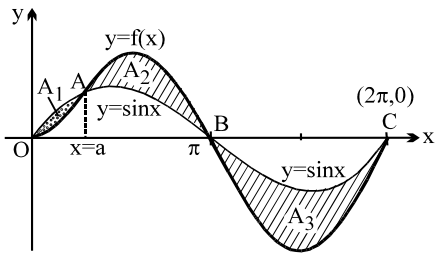
$$n = \frac{2 \pm \sqrt{4+4}}{2}$$

$n = 1 \pm \sqrt{2} \qquad \Rightarrow n \neq 1 - \sqrt{2}\,,\ n = 1 + \sqrt{2}$

10. In the adjacent figure, graphs of two functions y = f(x) and y = sinx are given. y = sinx intersects, y = f(x) at A(a, f(a)); B(π, 0) and C(2π, 0). A_i (i = 1, 2, 3,) is the area bounded by the curves y = f(x) and y = sinx between x=0 and x= a; i = 1, between x = a and x = π; i = 2, between x = π and x = 2π; i = 3. If $A_1 = 1 - \sin a + (a-1)\cos a$, determine the function f(x). Hence determine 'a' and A_1. Also calculate A_2 and A_3.



दिया गया चित्र फलन y = f(x) और y = sin x को दर्शाता है। यदि y = sin x, y = f(x) को A(a, f(a)); B(π, 0) और C(2π, 0) पर प्रतिच्छेदित करता है तथा A_i (i = 1, 2, 3,) वक्रों y = f(x) और y = sin x, x = 0 और x = a; i = 1, x = a और x = π; i = 2, और x = π और x = 2π; i = 3 से क्रमशः परिबद्ध क्षेत्रफल है। यदि $A_1 = 1 - \sin a + (a-1)\cos a$, तथा फलन f(x) ज्ञात कीजिए और a का मान बतलाईये, साथ ही A_2 और A_3 ज्ञात कीजिए।



Ans. $f(x) = x\sin x,\ a = 1;\ A_1 = 1 - \sin 1;\ A_2 = \pi - 1 - \sin 1;\ A_3 = (3\pi - 2)$ sq. units

Sol. $A_1 = \int_0^a \left(\sin x - f(x)\right)\cdot dx$ _____(i)

$1 - \sin A + (a-1)\cos a = -(\cos x)_0^a - \int_0^a f(x)\cdot dx$

$\int_0^a f(x)\cdot dx = -\cos a + 1 - 1 + \sin a - (a-1)\cos a$

$\int_0^a f(x)\cdot dx = \sin a - a\cos a$

By Newton Leibnitz Rule

$f(a) = \cos a - \left[a(-\sin a) + \cos a\right]$

$f(a) = \cos a + a\sin a - \cos a$

$\Rightarrow f(a) = a\sin a$

$\therefore f(x) = x\sin x$ _____(ii)

from $y = \sin x$ and $y = x\sin x$ point of intersection is "a"

$x\sin x = \sin x$

$(x-1)\sin x = 0$

$\Rightarrow \sin x = n\pi$ or $x = 1$

Hence $a = 1$

Now $A_1 = 1 - \sin 1$

$A_2 = \int_1^{\pi} (x\sin x - \sin x)\,dx$

$$A_2 = \int_1^{\pi} (x-1)\sin x \cdot dx$$

$$A_2 = \left(-(x-1)\cos x\right)_1^{\pi} + \int_1^{\pi} \cos x \cdot dx$$

$$A_2 = -\left((x-1)\cos x\right)_1^{\pi} + \sin\pi - \sin 1$$

$$A_2 = -\left[(\pi-1)\cos\pi\right] - \sin 1$$

$$A_2 = \pi - 1 - \sin 1$$

Again $$A_3 = \int_{\pi}^{2\pi} \left|x\sin x - \sin x\right| \cdot dx$$

$$A_3 = 3\pi - 2$$

11. Consider the two curves $y = 1/x^2$ and $y = 1/[4(x-1)]$.

(i) At what value of 'a' $(a > 2)$ is the reciprocal of the area of the fig. bounded by the curves, the lines $x = 2$ and $x = a$ equal to 'a' itself?

(ii) At what value of 'b' $(1 < b < 2)$ the area of the figure bounded by these curves, the lines $x = b$ and $x = 2$ equal to $1 - 1/b$.

माना दो वक्र $y = 1/x^2$ और $y = 1/[4(x-1)]$ दिये गये हैं।

(i) ५ ६; झ 2द्धके किस मान के लिये दिये गये वक्रों और रेखाओं $x = 2$ व $x = a$ से परिबद्ध क्षेत्र के क्षेत्रफल का प्रतिलोम भी 'a' स्वयं है ?

(ii) ५झ६ ;1 ढ इ ढ 2द्धके किस मान के लिये दिये गये वक्रों तथा रेखाओं $x = b$ और $x = 2$ से परिबद्ध क्षेत्र का क्षेत्रफल $1 - 1/b$ होगा।

Ans. $a = 1 + e^2$, $b = 1 + e^{-2}$

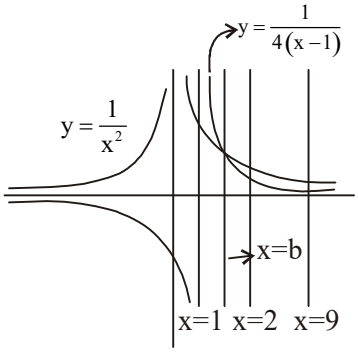
Sol. $y = \dfrac{1}{x^2}$ $\qquad$ $y = \dfrac{1}{4(x-1)}$

(i) Point of intersection

$$\frac{1}{x^2} = \frac{1}{4(x-1)}$$

$$x^2 - 4x + 4 = 0$$

$$x = 2$$



$$A = \int_2^a \left(\frac{1}{4(x-1)} - \frac{1}{x^2} \right) dx$$

$$= \frac{1}{4}\log(x-1) + \frac{1}{x}\Big|_2^a$$

$$= \frac{1}{4}\log(a-1) + \frac{1}{a} - \frac{1}{2}$$

Given $A = \frac{1}{a}$

$$\frac{1}{4}\log(a-1) + \frac{1}{a} - \frac{1}{2} = \frac{1}{a}$$

$a - 1 = e^2 \Rightarrow a = 1 + e^2$

(ii) $A = \int_b^2 \frac{1}{4(n-1)} - \left(\frac{1}{x^2} \right) dx$

$$= \frac{1}{4}\log(x-1) + \frac{1}{2}\Big|_b^2$$

$$= +\frac{1}{2} - \frac{1}{b} - \frac{1}{4}\log(b-1)$$

Given $A = 1 - \frac{1}{b}$

$$+\frac{1}{2} - \frac{1}{b} - \frac{1}{4}\log(b-1) = 1 - \frac{1}{b}$$

$$-\frac{1}{4}\log(b-1) = \frac{1}{2}$$

$\log(b-1) = -2 \Rightarrow b = 1 + e^{-2}$

12. Show that the area bounded by the curve $y = \frac{\ell n\, x - c}{x}$, the x-axis and the vertical line through the maximum point of the curve is independent of the constant c.

सिद्ध कीजिये कि वक्र $y = \frac{\ell n\, x - c}{x}$, x-अक्ष और वक्र के उच्चिष्ठ बिन्दु से खींची गयी उर्ध्वाधर रेखा से परिबद्ध क्षेत्र का क्षेत्रफल अचर 'c' से स्वतंत्र है।

Ans. 1/2

Sol. $y = \frac{\ell nx - C}{x}$

at $y = 0_{x\,=\,e^{c}}$

$$\frac{dy}{dx} = \frac{1-(\ell nx - c)}{x^2} = 0$$

$$\Rightarrow x = e^{1+c}$$

$$A = \int_{e^c}^{e^{1+c}} \left(\frac{\ell nx - c}{x}\right) dx$$

$$A = \frac{(\ell nx - c)^2}{2}\Bigg|_{e^c}^{e^{1+c}}$$

$$A = \frac{1}{2}$$

13. For what value of 'a' is the area of the figure bounded by the lines, $y = \frac{1}{x}$, $y = \frac{1}{2x-1}$, $x = 2$ and $x = a$, $a > 0$ equal to $\ell n \frac{4}{\sqrt{5}}$?

'a' के किस मान के लिये वक्रों, $y = \frac{1}{x}$, $y = \frac{1}{2x-1}$, $x = 2$ और $x = a$, $a > 0$ से परिबद्ध क्षेत्रफल $\ell n \frac{4}{\sqrt{5}}$ होगा?

Ans. $a = 8$ or $\frac{2}{5}\left(6 - \sqrt{21}\right)$

Sol. POI $y = \frac{1}{x}$; $y = \frac{1}{2x-1}$; $x = 2$ & $x = a$

$x = 2x - 1$

x = 1

Case-I : a > 2

$$A = \int_2^a \left(\frac{1}{x} - \frac{1}{2x-1}\right) dx$$

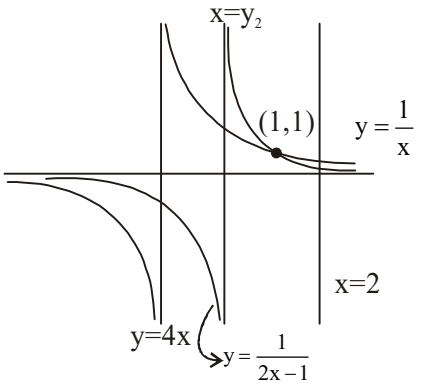
$$\ell n \frac{4}{\sqrt{5}} = \ell nx - \frac{1}{2}\ell n(2x-1)\Big|_2^a$$

$$= \ell na - \frac{1}{2}\ell n(2a-1) - \ell n2 + \frac{1}{2}\ell n3$$

$$\ell n \frac{4}{\sqrt{5}} = \ell n\frac{a}{2} - \frac{1}{2}\ell n\frac{2a-1}{3}$$

$$\frac{1}{2}\ell n\left(\frac{2a-1}{3}\right) = \ell n\frac{a}{2} \times \frac{\sqrt{5}}{4}$$

$$\frac{2a-1}{3} = \frac{a^2 \times 5}{64}$$



$128a - 64 = 15a^2$

$15a^2 - 128a + 64 = 0$

$15a^2 - 120a - 8a + 64 = 0$

$$(15a - 8)(a - 8) = 0 \Rightarrow a = \frac{8}{15}, 8$$

a = 8

Case-II :

1 < a < 2

$$A = \int_a^2 \left(\frac{1}{x} - \frac{1}{2x-1}\right) dx$$

$$\ell n \frac{4}{\sqrt{5}} = \left(\ell nx - \frac{1}{2}\ell n(2x-1)\right)\Big|_a^2$$

$$\ell n \frac{4}{\sqrt{5}} = \frac{1}{2}\ell n\left(\frac{2a-1}{3}\right) - \ell n\frac{a}{2}$$

$$\ell n\frac{2a}{\sqrt{5}} = \frac{1}{2}\ell n\left(\frac{2a-1}{3}\right)$$

$$\frac{4a^2}{5} = \frac{2a-1}{3}$$

$12a^2 = 10a - 5$

$12a^2 - 10a + 5 = 0$

$D = 100 - 240 < 0$

Case-III :

$$\frac{1}{2} < a < 1$$

$$A = \int_a^1\left(\frac{1}{2x-1} - \frac{1}{x}\right)dx + \int_1^2\left(\frac{1}{x} - \frac{1}{2x-1}\right)dx$$

$$\ell n\frac{4}{\sqrt{5}} = \frac{1}{2}\ell n(2x-1) - \ell nx\Big|_a^1 + \left(\ell nx - \frac{1}{2}\ell n(2x-1)\right)\Big|_1^2$$

$$\ell n\frac{4}{\sqrt{5}} = \frac{1}{2}\ell n\frac{a^2}{2a-1} + \frac{1}{2}\ell n\left(\frac{4}{3}\right)$$

$$\frac{16}{5} = \left(\frac{a^2}{2a-1}\right)\frac{4}{3}$$

$12(2a - 1) = 5a^2$

$24a - 12 = 5a^2$

$5a^2 - 24a + 12 = 0$

$$a = \frac{24 \pm \sqrt{24^2 - 4\times 12\times 5}}{10}$$

$$a = \frac{24 \pm \sqrt{12\times 4\times 7}}{10}$$

$$a = \frac{12 \pm 2\sqrt{21}}{5}$$

$a = \frac{2}{5}\left(6+\sqrt{21}\right), \frac{2}{5}\left(6-\sqrt{21}\right)$

$a \neq \frac{2}{5}\left(6+\sqrt{21}\right)$

so, $a = \frac{2}{5}\left(6-\sqrt{21}\right)$

14. For the curve $f(x) = \frac{1}{1+x^2}$, let two points on it are $A\left(\alpha, f(\alpha)\right)$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$ $(\alpha > 0)$. Find the minimum area bounded by the line segments OA, OB and $f(x)$, where 'O' is the origin.

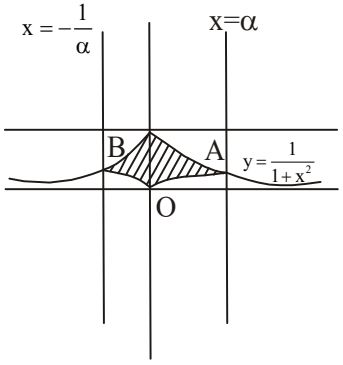
माना $A\left(\alpha, f(\alpha)\right)$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$ $(\alpha > 0)$ वक्र $f(x) = \frac{1}{1+x^2}$ पर दो बिन्दु है तथा रेखाओं OA, OB और वक्र $f(x)$, द्वारा परिबद्ध न्यूनतम क्षेत्रफल ज्ञात कीजिये। (जहाँ 'O' मूलबिन्दु है)

Ans. $\frac{(\pi-1)}{2}$

Sol. $f(\alpha) = \frac{1}{1+\alpha^2}$

$f\left(-\frac{1}{\alpha}\right) = \frac{\alpha^2}{1+\alpha^2}$

$A = \int_{-\frac{1}{\alpha}}^{\alpha} \frac{1}{1+x^2} - \frac{1}{2}\times\frac{1}{\alpha}\times\frac{\alpha^2}{1+\alpha^2} - \frac{1}{2}\times\alpha\times\frac{1}{1+\alpha^2}$



$A = \tan^{-1}(x)\Big|_{-\frac{1}{\alpha}}^{\alpha} - \frac{1}{2}\frac{\alpha}{1+\alpha^2} - \frac{1}{2}\frac{\alpha}{1+\alpha^2}$

$A = \tan^{-1}\alpha + \tan^{-1}\left(\frac{1}{\alpha}\right) - \frac{\alpha}{1+\alpha^2}$

$A = \tan^{-1}\alpha + \cot^{-1}\alpha - \frac{\alpha}{1+\alpha^2}$

$$A = \frac{\pi}{2} - \frac{\alpha}{1+\alpha^2}$$

$$\frac{dA}{d\alpha} = -\frac{\left(1+\alpha^2\right) - 2\alpha^2}{\left(1+\alpha^2\right)} = 0$$

$1 - \alpha^2 = 0$

$\alpha = 1 \qquad \alpha > 0$

$$A = \frac{\pi}{2} - \frac{1}{2}$$

$$A = \frac{\pi - 1}{2}$$

15. Let 'c' be the constant number such that $c > 1$. If the least area of the figure given by the line passing through the point (1, c) with gradient '*m*' and the parabola $y = x^2$ is 36 sq. units find the value of $(c^2 + m^2)$.

माना 'c' अचर संख्या इस प्रकार है कि $c > 1$. यदि बिन्दु (1, c) से जाने वाली तथा '*m*' प्रवणता की रेखा और परवलय $y = x^2$ से परिबद्ध न्यूनतम क्षेत्रफल 36 वर्ग इकाई है, तो $(c^2 + m^2)$ का मान ज्ञात कीजिये।

Ans. 104

Sol. Let line $y - c = m(x - 1)$

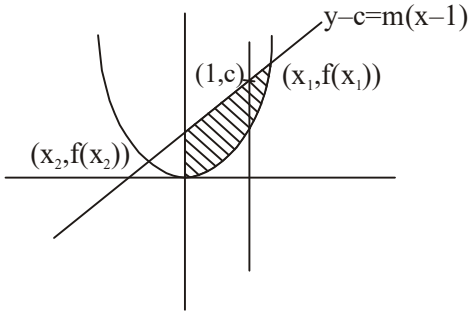
POI

$y = mx + c - m$

& $y = x^2$

$x^2 = mx + c - m$

$$x^2 - mx + m - c = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$



$x_1 + x_2 = m$

$x_1 x_2 = m - c$

$$x_2 - x_1 = \sqrt{m^2 - 4m + 4c}$$

$$x_2^2 - x_1^2 = m\sqrt{m^2 - 4m + 4c}$$

$$x_2^3 - x_1^3 = (x_2 - x_1)\left(x_2^2 + x_1^2 + x_2x_1\right)$$

$$= (x_2 - x_1)\left((x_1 + x_2)^2 - x_1x_2\right)$$

$$= \sqrt{m^2 - 4m + 4c}\left(m^2 - m + c\right)$$

$$A = \int_{x_1}^{x_2}\left(mx + c - m - x^2\right)dx$$

$$A = \frac{m}{2}x^2 + (c - m)x - \frac{x^3}{3}\Bigg|_{x_1}^{x_2}$$

$$A = \frac{m}{2}\left(x_2^2 - x_1^2\right) + (c - m)(x_2 - x_1) - \frac{1}{3}\left(x_2^3 - x_1^3\right)$$

$$A = (x_2 - x_1)\left(\frac{m}{2}(x_2 + x_1) + (c - m) - \frac{1}{3}\left(x_1^2 + x_2^2 + x_1x_2\right)\right)$$

$$A = \sqrt{m^2 - 4m + 4c}\left(\frac{m^2}{2} + c - m - \frac{1}{3}\left(m^2 - m + c\right)\right)$$

$$A = \sqrt{m^2 - 4m + 4c}\left(\frac{m^2}{6} - \frac{2}{3}m + \frac{2}{3}c\right)$$

for minimum area

$$\frac{dA}{dm} = 0$$

$$\sqrt{m^2 - 4m + 4c}\left(\frac{m}{3} - \frac{2}{3}\right) + \left(\frac{m^2 - 4m + 4c}{6}\right)\frac{2m - 4}{\sqrt{m^2 - 4m + 4c}} = 0$$

$$\left(m^2 - 4m + 4c\right)\left(\frac{m - 2}{3}\right) + \frac{\left(m^2 - 4m + 4c\right)}{3}(m - 2) = 0$$

$m = 2$ or $m^2 - 4m + 4c = 0$ (not possible)

so, $m = 2$

$$A_{min.} = \sqrt{4c-4}\left(\frac{4}{6}-\frac{4}{3}+\frac{2}{3}c\right)$$

$$36 = 2\sqrt{c-1}\left(\frac{2}{3}c-\frac{2}{3}\right)$$

$$36 = \frac{4}{3}(c-1)^{\frac{3}{2}}$$

$$(c-1)^{\frac{3}{2}} = 27 \Rightarrow (c-1)^{\frac{1}{2}} = 3$$

$c - 1 = 9 \Rightarrow c = 10$

$c^2 + m^2 = 100 + 4 = 104$

16. Let A_n be the area bounded by the curve $y = (\tan x)^n$ & the lines $x = 0$, $y = 0$ & $x = \pi/4$. Prove that for $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ & deduce that $1/(2n+2) < A_n < 1/(2n-2)$.

यदि A_n वक्र $y = (\tan x)^n$ तथा सरल रेखाओं $x = 0$, $y = 0$ और $x = \pi/4$ से परिबद्ध क्षेत्रफल है, तो सिद्ध कीजिए $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ तथा $1/(2n+2) < A_n < 1/(2n-2)$.

Ans.

Sol. $A_n = \int_0^{\pi/4} (\tan x)^n \, dx$

$$A_n + A_{n-2} = \int_0^{\pi/4} \left[(\tan x)^n + (\tan x)^{n-2}\right] dx$$

$$= \int_0^{\pi/4} (\tan x)^{n-2} \sec^2 x \, dx = \left[\frac{t^{n-1}}{n-1}\right]_0^1 = \frac{1}{n-1}$$

Also $A_{n+2} < A_n < A_{n-2}$

$$\Rightarrow \frac{1}{n+1} < 2A_n < \frac{1}{n-1}$$

17. Consider the two curves $C_1 : y = 1 + \cos x$ & $C_2 : y = 1 + \cos(x-\alpha)$ for $\alpha \in (0, \pi/2)$; $x \in [0, \pi]$. Find the value of α, for which the area of the figure bounded by the curves C_1, C_2 & $x = 0$ is same as that of the figure bounded by C_2, $y = 1$ & $x = \pi$. For this value of α, find the ratio in which the line $y = 1$ divides the area of the figure by the curves C_1, C_2 & $x = \pi$.

दिया है $C_1 : y = 1 + \cos x$ और $C_2 : y = 1 + \cos(x-\alpha)$ for $\alpha \in (0, \pi/2)$ और $x \in [0, \pi]$ दो वक्र हैं। α का मान ज्ञात कीजिये यदि वक्रो C_1, C_2 और $x = 0$ से परिबद्ध क्षेत्रफल और C_2, $y = 1$ और $x = \pi$ से परिबद्ध क्षेत्रफल समान है। इस α के मान के लिये वह अनुपात ज्ञात कीजिये जबकि वक्रों C_1, C_2 और $x = \pi$ से परिबद्ध क्षेत्रफल को रेखा $y = 1$ विभाजित करती है।

Ans. $\alpha = \pi/3$, ratio $= 2 : \sqrt{3}$

Sol. $C_1 : y = 1 + \cos x$

$C_2 : y = 1 + \cos(x - \alpha) \qquad \alpha \in \left(0, \frac{\pi}{2}\right),\ x \in [0, \pi]$

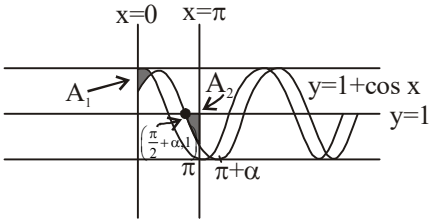
POI of C_1 & C_2

$\cos x = \cos(x - \alpha)$

$x = |x - \alpha|$

$2\xi = \alpha$

$x = \frac{\alpha}{2}$



Area

$A_1 : C_1, C_2$ & $x = 0$

$A_2 : C_2, y = 1$ & $x = \pi$

$A_1 = A_2$

$$\int_0^{\frac{\alpha}{2}} \left(\cos x - \cos(x-\alpha)\right) dx = \int_{\frac{\pi}{2}+\alpha}^{\pi} -\cos(x-\alpha)\, dx$$

$$\sin x - \sin(x-\alpha)\Big|_0^{\frac{\alpha}{2}} = \sin(x-\alpha)\Big|_{\pi}^{\frac{\pi}{2}+\alpha}$$

$$\sin\frac{\alpha}{2} + \sin\frac{\alpha}{2} - \sin\alpha = 1 - \sin\alpha$$

$$\sin\frac{\alpha}{2} = \frac{1}{2}$$

$$\frac{\alpha}{2} = \frac{\pi}{6}$$

$$\alpha = \frac{\pi}{3}$$

18. For what values of $a \in [0,1]$ does the area of the figure bounded by the graph of the function $y = f(x)$ and the straight lines $x = 0$, $x = 1$ & $y = f(a)$ is at a minimum & for what values it is at a maximum if $f(x) = \sqrt{1-x^2}$. Find also the maximum & the minimum areas.

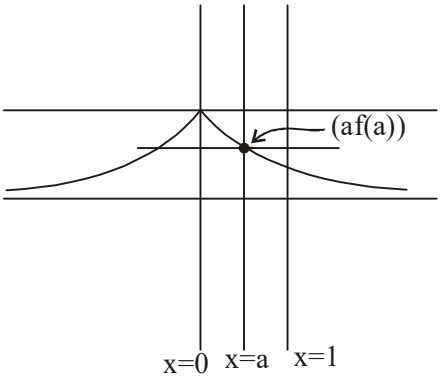
$\in [0,1]$ के किस मान के लिये वक्र $y = f(x)$ सरल रेखाऐं $x = 0$, $x = 1$ और $y = f(a)$ से परिबद्ध क्षेत्रफल न्यूनतम होगा तथा किस मान के लिये यह अधिकतम होगा यदि $f(x) = \sqrt{1-x^2}$ साथ ही अधिकतम और न्यूनतम क्षेत्रफल भी ज्ञात कीजिये।

Ans. $a = 1/2$ gives minima, $A\left(\frac{1}{2}\right) = \frac{3\sqrt{3}-\pi}{12}$; $a = 0$ gives local maxima $A(0) = 1 - \frac{\pi}{4}$; $a = 1$ gives maximum value, $A(1) = \pi/4$

Sol. Given $a \in [0,1]$

$$A = \int_0^a \left(\sqrt{1-x^2} - f(a)\right)dx + \int_a^1 \left(f(a) - \sqrt{1-x^2}\right)dx$$

$$A = \int_0^a \sqrt{1-x^2}\,dx - \int_0^a f(a)\,dx + \int_a^1 f(a)\,dx - \int_0^1 \sqrt{1-x^2}\,dx$$



$$\left[\frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\sin^{-1}(x)\right]_0^a - \left[\frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\sin^{-1}(x)\right]_a^1$$

$-a\,f(0) + (1-a)\,f(a)$

$$= \frac{a}{2}\sqrt{1-a^2} - \frac{1}{2}\sin^{-1}a + \frac{1}{2}\times\frac{\pi}{2} + \frac{a}{2}\sqrt{1\cdot a^2} - \frac{1}{2}\sin^{-1}a$$

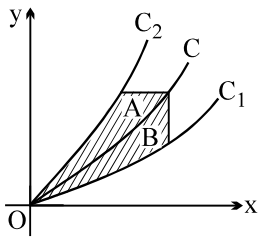
$$-a + (1-a)\sqrt{1-a^2}$$

$$A = \sqrt{1-a^2} - \sin^{-1}a + \frac{\pi}{4}$$

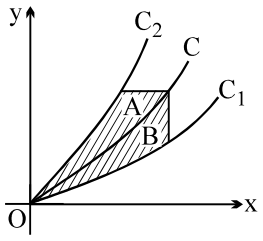
$$\frac{dA}{da} = \frac{-2a}{\sqrt{1-a^2}} - \frac{1}{\sqrt{1-a^2}} = 0$$

$$a = -\frac{1}{2}$$

19. Let C_1 and C_2 be two curves passing through the origin as shown in the figure. A curve C is said to "bisect the area" the region between C_1 and C_2, if for each point P of C, the two shaded regions A and B shown in the figure have equal areas. Determine the upper curve C_2, given that the bisecting curve C has the equation $y = x^2$ & that the lower curve C_1 has the equation $y = x^2/2$.



माना C_1 और C_2 मूल बिन्दु से जाने वाले वक्र चित्रनुसार है। यदि वक्र C, C_1 और C_2 के क्षेत्र को C पर प्रत्येक बिन्दु P के लिये समद्विभाजित करता है, तथा चित्र में दो परिबद्ध क्षेत्रफल A और B समान क्षेत्रफल वाले दिये गये हैं। तब ऊपरी वक्र C_2 ज्ञात कीजिये। जबकि वक्र C और C_1 के समीकरण क्रमशः $y = x^2$ और $y = x^2/2$ दिये गये हैं।



Ans. $(16/9)\,x^2$

Sol. $C : y = x^2$

$C_1 : y = \dfrac{x^2}{2}$

Let $C_2 : y = f(x)$

Let point of curve C is (α, α^2)

$A_1 = B$

$$\int_0^{\alpha^2} (C - C_2)\,dy = \int_0^{\alpha} (C - C_1)\,dx$$

$$\int_0^{\alpha^2} \left(\sqrt{y} - f^{-1}(y)\right)dy = \int_0^{\alpha} \left(x^2 - \frac{x^2}{2}\right)dx$$

$$\left(\alpha - f^{-1}\left(\alpha^2\right)\right)2\alpha = \frac{\alpha^2}{2}$$

$$\alpha - f^{-1}\left(\alpha^2\right) = \frac{\alpha}{4}$$

$$f^{-1}\left(\alpha^2\right) = \frac{3}{4}\alpha$$

$$f\left(\frac{3}{4}\alpha\right)=\alpha^2$$

$$f(x)=\frac{16}{9}x^2$$

$$\because \frac{3}{4}\alpha=x$$

$$\alpha=\frac{4}{3}x$$

$$\therefore\ y=\alpha^2=\frac{16}{9}x^2$$

20. Given f (x) = $\int_0^x e^t(\ln\sec t-\sec^2 t)\,dt$; g (x) = – $2e^x$ tan x. Find the area bounded by the curves $y=f(x)$ and $y=g(x)$ between the ordinates $x=0$ and $x=\frac{\pi}{3}$.

माना $f(x)=\int_0^x e^t(\ln\sec t-\sec^2 t)\,dt$; $g(x)=-2e^x\tan x$ तब $y=f(x)$, $y=g(x)$ और $x=0$ तथा $x=\frac{\pi}{3}$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

Ans. $e^{\pi/3}\log 2$ sq. units

Sol. $f(x)=\int_0^x e^t\left(\ell n\sec t-\sec^2 t\right)$

$$=\int_0^x e^t\left(\ell n\sec t+\tan t\right)-e^t\left(\tan t+\sec^2 t\right)dt$$

$$=e^t\left(\ell n\sec t-\tan t\right)\Big|_0^x=e^x\left(\ell n\sec x-\tan x\right)$$

Area between f(x) & g(x)

$$A=\int_0^{\frac{\pi}{3}}\left(f(x)-g(x)\right)dx$$

$$= \int_0^{\frac{\pi}{3}} e^x \left(\ell n \sec x + \tan x\right) dx$$

$$= e^x \ell n \sec x \Big|_0^{\frac{\pi}{3}}$$

$$= e^{\frac{\pi}{3}} \ell n 2$$

21. Let $f: [0, \infty) \to R$ be a continuous and strictly increasing function such that $f^3(x) = \int_0^x t f^2(t)\, dt$, $\forall\ x > 0$. Find the area enclosed by $y = f(x)$, the x-axis and the ordinate at $x = 3$.

माना $f : [0, \infty) \to R$ सतत् तथा निरतंर वर्धमान फलन इस प्रकार है कि $f^3(x) = \int_0^x t f^2(t)\, dt$, $\forall\ x > 0$, तब $y = f(x)$, x-अक्ष और $x = 3$ से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. 3/2

Sol. $f^3(x) = \int_0^x t f^2(t)\, dt$

$f(0) = 0$

$f \Rightarrow$ strictly increasing

$3f^2(x)\, f'(x) = x f^2(x)$

$$f'(x) = \frac{x}{3}$$

$$f(x) = \frac{x^2}{6} + C$$

$f(0) = C = 0$

$$f(x) = \frac{x^2}{6}$$

$$A = \int_0^3 \frac{x^2}{6} dx$$

$$= \frac{x^3}{18}\Big|_0^3 = \frac{3}{2}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 22 to 24

If f is smooth on [a, b], the length of the curve y = f(x) from a to b is $L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b \sqrt{1+(f'(x))^2}\,dx$ .....(i)

At point on a curve where dy/dx fails to exist, dx/dy may exist and we may be able to find the curve 's length by expressing x as a function of y and applying the following analogue of Eq.(i) Formula for the length of a smooth

curve $x = g(y),\ c \le y \le d$ $L = \int_c^b \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy = \int_c^d \sqrt{1+(g'(y))^2}\,dy$ .....(ii)

यदि f अन्तराल [a, b] में अवकलनीय है तथा a से b में वक्र y = f(x) की लम्बाई निम्न प्रकार से परिभाषित की जाती है

$$L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b \sqrt{1+(f'(x))^2}\,dx \qquad \text{.....(i)}$$

यदि वक्र के किसी बिन्दु पर dy/dx विद्यमान नहीं है तथा dx/dy विद्यमान हो सकता है तो हम वक्र की लम्बाई x को y के फलन के रूप esa O;Dr djds fuEu iz dkj ls Kkr dj l drs gSA Eq.(i) की मदद से वक्र की लम्बाई $x = g(y),\ c \le y \le d$

$$L = \int_c^b \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy = \int_c^d \sqrt{1+(g'(y))^2}\,dy \qquad \text{.....(ii)}$$

22. The length of the curve $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1,\ 0 \le x \le 1$ is :

(A) $\frac{13}{6}$ (B) 2 (C) $\frac{1}{12}$ (D) none of these

वक्र $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1,\ 0 \le x \le 1$ की लम्बाई होगी :

(A) $\frac{13}{6}$ (B) 2 (C) $\frac{1}{12}$ (D) कोई नहीं

Ans. A

Sol. $y = \frac{4\sqrt{2}}{3}x^{\frac{3}{2}} - 1 \qquad 0 \le x \le 1$

$$\frac{dy}{dx} = 2\sqrt{2}x^{\frac{1}{2}}$$

$$L = \int_0^1 \sqrt{1+8x}\,dx$$

$$= \frac{2}{3}\frac{(1+8x)^{\frac{3}{2}}}{8}\Bigg|_0^1 = \frac{2}{3}\times\frac{27}{8} - \frac{1}{12}$$

$$= \frac{27}{12} - \frac{1}{12}$$

$$= \frac{13}{6}$$

23. The length of the curve $y = (x/2)^{2/3}$ from $x = 0$ to $x = 2$ is :

(A) 3.37 (B) 2.27 (C) 2.0 (D) none of these

वक्र $y = (x/2)^{2/3}$ की $x = 0$ तथा $x = 2$ के मध्य की लम्बाई होगी :

(A) 3.37 (B) 2.27 (C) 2.0 (D) कोई नहीं

Ans. B

Sol. $y = \left(\frac{x}{2}\right)^{\frac{2}{3}} \Rightarrow x = 0 \text{ to } y = 2$

$$\frac{dy}{dx} = \frac{2}{3\left(\frac{x}{2}\right)^{\frac{1}{3}}} \qquad x = 0 \text{ not defined}$$

$x = 2y^{\frac{3}{2}} \Rightarrow y = 0 \text{ to } y = 1$

$$\frac{dx}{dy} = 3y^{\frac{1}{2}}$$

$$L = \int_0^1 \sqrt{1+9y}\,dy$$

$$= \frac{2}{3}\frac{(1+9y)^{\frac{3}{2}}}{9}\Bigg|_0^1$$

$$= \frac{2}{3}\times\frac{10^{\frac{3}{2}}}{9} - \frac{2}{3\times 9}$$

$$= \frac{2\times 31{\cdot}62}{27} - \frac{2}{27}$$

$= 2.266$

24. The length of the curve $y = (1/3)\,(x^2+2)^{3/2}$ from $x = 0$ to $x = 3$ is :

(A) 10 (B) 12 (C) $\frac{1}{12}$ (D) none of these

वक्र $y = (1/3)\,(x^2+2)^{3/2}$ की $x = 0$ से $x = 3$ के मध्य की लम्बाई होगी :

(A) 10 (B) 12 (C) $\frac{1}{12}$ (D) कोई नही

Ans. B

Sol. $y = \frac{1}{3}\left(x^2+2\right)^{\frac{3}{2}}$ $x = 0$ to $x = 3$

$$\frac{dy}{dx} = \frac{1}{2}\left(x^2+2\right)^{\frac{1}{2}}\cdot 2x$$

$$L = \int_0^3 \sqrt{1+\frac{1}{4}\left(x^2+2\right)\cdot 4x^2}\,dx$$

$$L = \int_0^3 \sqrt{1+x^4+2x^2}\,dx$$

$$= \int_0^3 \left(x^2+1\right)dx$$

$$= \frac{x^3}{3} + x \Big|_0^3$$

$= 9 + 3 = 12$

Paragraph for Question nos. 25 to 28

Let $f(x)$ be a differntiable function such that $f(x+y) = e^x f(y) + e^y f(x)$ all x,y and $f'(0) = 1$

माना $f(x)$ अवकलनीय फलन इस प्रकार है कि $f(x+y) = e^x f(y) + e^x f(x)$ सभी x,y के लिये तथा $f'(0) = 1$

25. f(x) has :

(A) maximum (B) minimum

(C) both maximum and minimum (D) neither maximum nor minimum

f(x) रखता है :

(A) उच्चिष्ठ (B) निम्निष्ठ

(C) उच्चिष्ठ तथा निम्निष्ठ (D) ना तो उच्चिष्ठ न ही निम्निष्ठ

Ans. B

Sol. $f(x + y) = e^x f(y) + e^y f(x)$

$f'(0) = 1 \qquad f(0) = 0$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{e^x f(h) + e^h f(x) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{e^x f'(h) + e^h f(x)}{h}$$

$$= e^x f'(0) + e^0 f(x)$$

$$f'(x) = e^x + f(x)$$

$$\frac{dy}{dx} - y = e^x$$

I.F. $= e^{-x}$

$$y \cdot e^{-x} = \int dx$$

$y = (x + c)e^x$

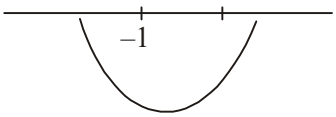
$x = 0, y = 0$

$0 = c \Rightarrow c = 0$

$y = x e^x$

$$\frac{dy}{dx} = e^x (x+1) = 0$$

$x = -1$



minima

26. The range of f(x) is :

f(x) का परिसर होगा :

(A) R (B) $[0, \infty]$ (C) $\left(-\frac{1}{e}, 1\right)$ (D) $\left[-\frac{1}{e}, \infty\right)$

Ans. D

Sol. $f_{min}(-1) = -\frac{1}{e}$

Range $\left[-\frac{1}{e}, \infty\right)$

27. $\lim\limits_{x \to -\infty} f(x)$ is :

(A) 0 (B) 1 (C) –1 (D) non existent

$\lim\limits_{x \to -\infty} f(x)$ होगा :

(A) 0 (B) 1 (C) –1 (D) विद्यमान नहीं है

Ans. A

Sol. $\lim\limits_{x \to -\infty} f(x) = \lim\limits_{x \to -\infty} xe^x$

$= \lim\limits_{x \to -\infty} \frac{x}{e^{-x}}$

$= \lim_{x \to -\infty} \frac{1}{-e^{-x}}$

$= 0$

28. The area bounded by the curve $y = f(x)$ and the x–axis is :

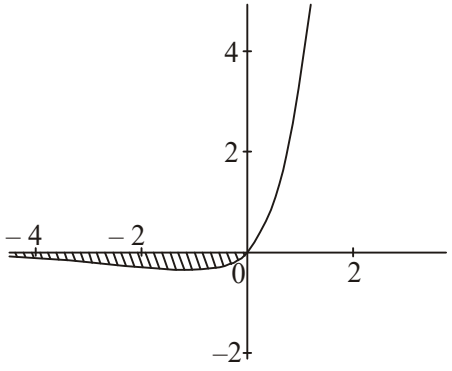
वक्र $y = f(x)$ तथा x–अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 1 (B) $\frac{1}{2}$ (C) 2 (D) e

Ans. A

Sol. Required area $= \int_{-\infty}^{0} \left| x \cdot e^x \right| dx = -\left(e^x (x-1) \right)_{-\infty}^{0} = -\left(e^0 (0-1) - 0 \right)$

$= 1$ sq. units



MATCH THE COLUMN TYPE QUESTION(S)

29. Match the following column :

Column I	**Column II**
(A) The area between the parabolas $y^2 = 4ax$ and $x^2 = 8ay$ is	(P) 27
(B) Area bounded by the curve $y = (x-1)(x-2)(x-3)$ and x-axis lying between the ordinates $x = 0$ and $x = 3$ is equal to	(Q) 2π
(C) The area cut off the parabola $4y = 3x^2$ by the straight line $2y = 3x + 12$ in sq. units is	(R) $\frac{32}{3}a^2$
(D) The area of the region satisfying $\lvert x \rvert + \lvert y \rvert \le \sqrt{\pi}$ is	(S) $\frac{11}{4}$

निम्न कॉलम को सुमेलित कीजिए :

Column I	**Column II**
(A) परवलयों $y^2 = 4ax$ तथा $x^2 = 8ay$ के मध्य का क्षेत्रफल होगा	(P) 27
(B) वक्र $y = (x - 1)(x - 2)(x - 3)$, x-अक्ष तथा रेखाओं $x = 0$ और $x = 3$ के मध्य क्षेत्रफल होगा	(Q) 2π
(C) परवलय $4y = 3x^2$ तथा सरल रेखा $2y = 3x + 12$ के मध्य का क्षेत्रफल होगा	(R) $\frac{32}{3}a^2$
(D) वक्र $\lvert x\rvert + \lvert y\rvert \le \sqrt{\pi}$ से घिरा हुआ क्षेत्रफल होगा	(S) $\frac{11}{4}$

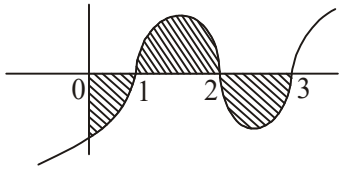
Ans. A - R, B - S, C - P, D - Q

Sol. (A) $y^2 = 4ax$

$x^2 = 8ay$

$$A = \frac{16}{3} \times a \times b$$

$$A = \frac{32}{3}a^2$$

(B) 

$$A = -\int_0^1 (x-1)(x-2)(x-3) + \int_1^2 (x-1)(x-2)(x-3) - \int_2^3 (x-1)(x-2)(x-3)\,dx$$

$$\int (x-1)(x-2)(x-3)\,dx = \int \left(x^3 - 6x^2 + 11x - 6\right)dx$$

$$= \frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x$$

$$A = -\left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_0^1 + \left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_1^2 - \left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_2^3$$

$$A = \frac{9}{4} + \frac{1}{4} + \frac{1}{4} = \frac{11}{4}$$

(C) Point of Intersection

$4y = 3x^2$ & $2y = 3x + 12$

$3x^2 = 6x + 24$

$x^2 - 2x - 8 = 0$

$x = 4$ & $x = -2$

$y = 12$ $y = 3$

$$A = \int_{-2}^{4}\left(\frac{3x+12}{2} - \frac{3x^2}{4}\right)dx$$

$$= \left(\frac{3x^2}{4} + 6x - \frac{x^3}{4}\right)_{-2}^{4}$$

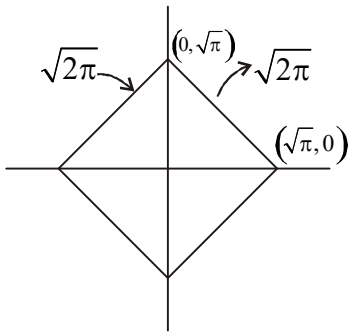
$= 20 - (-7)$

$= 27$

(D) $|x| + |y| \le \sqrt{\pi}$

$A = \sqrt{2\pi} \times \sqrt{2\pi}$

$= 2\pi$



30. Match the following column :

Column I	**Column II**
(A) The cosine of the angle between the curves $y = 3^{x-1} \ln x$ and $y = x^x - 1$ at their point of intersection on the line $y = 0$, is	(P) 0
(B) The area bounded by the curves $x = -4y^2$ and $(x - 1) = -5y^2$ is	(Q) 1
(C) The value of the integral $\int_0^{\pi/2} (\sin x)^{\cos x}(\cos x \cot x - \ln(\sin x)^{\sin x})\,dx$, is	(R) $\frac{4}{3}$

(D) A continuous function $f: [1, 6] \to [0, \infty)$ is such that $f'(x) = \frac{2}{x + f(x)}$ (S) $2\ ln\ 6$

and $f(1) = 0$, then the maximum value of f cannot exceed

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	Column I	**Column II**
(A)	वक्रों $y = 3^{x-1}\ ln\ x$ और $y = x^x - 1$ के मध्य रेखा $y = 0$ से प्रतिच्छेद बिन्दु पर कोज्या कोण का मान है	(P) 0
(B)	वक्रों $x = -4y^2$ और $(x - 1) = -5y^2$ से परिबद्ध क्षेत्रफल है	(Q) 1
(C)	समाकलन $\int_0^{\pi/2} (\sin x)^{\cos x} (\cos x \cot x - \ln(\sin x)^{\sin x})\, dx$ का मान है	(R) $\frac{4}{3}$
(D)	एक सतत् फलन $f: [1, 6] \to [0, \infty)$ इस प्रकार है कि $f'(x) = \frac{2}{x + f(x)}$ और $f(1) = 0$, तब f का मान अधिकतम मान से अधिक नहीं हो सकता	(S) $2\ ln\ 6$

Ans. (A) Q, (B) R, (C) Q, (D) S

Sol. (A) $y_1 = 3^{x-1} \ell nx \qquad y_2 = x^x - 1$

Point of intersection $y = 0$

$x = 1$

$$\frac{dy_1}{dx} = \frac{3^{x-1}}{x} + 3x^{x-1} \ell n3 \ell nx$$

$(1, 0)$

$m_1 = 1$

$$\frac{dy_2}{dx} = x^x (1 + \ell nx)$$

$(1,0)$

$m_2 = 1$

Angle $\theta = 0$

$\cos \theta = 1$

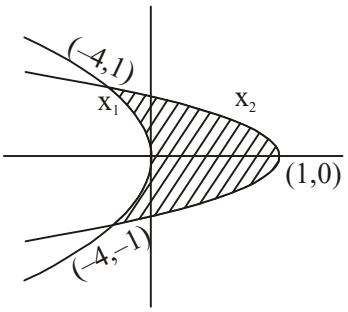
(B) $x = -4y^2 \qquad x - 1 = -5y^2$

Point of intersection $-4y^2 - 1 = -5y^2$

$y^2 = 1$

$y = \pm 1$

$$A = \int_{-1}^{1} (x_2 - x_1)\,dy$$



$$= \int_{-1}^{1} \left[1 - 5y^2 + 4y^2\right] dy$$

$$= \int_{-1}^{1} \left(1 - y^2\right) dy$$

$$= 2\int_{0}^{1} \left(1 - y^2\right) dy$$

$$= 2\left(y - \frac{y^3}{3}\right)_0^1 = \frac{4}{3}$$

(C) $$\int_{0}^{\frac{\pi}{2}} \sin x^{\cos x} \left(\cos x \cdot \cot x - \ell n\left(\sin x\right)^{\sin x}\right) dx$$

$$= \int_{0}^{\frac{\pi}{2}} \frac{d}{dx}\left(\sin x^{\cos x}\right) dx$$

$$= \left(\sin x\right)^{\cos x}\Big|_0^{\frac{\pi}{2}}$$

$= 1$

(D) $f'(x) > 0$ so f_{max} at $x = 6$

$$f'(x) = \frac{2}{x + f(x)}$$

$$\frac{dy}{dx} = \frac{2}{x + y}$$

$x + y = t$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx}-1=\frac{2}{t}$$

$$\frac{tdt}{2+t}=dx$$

$$\int\left(1-\frac{2}{2+t}\right)dt=\int dx$$

$$t-2\log(2+t)=x+C$$

$$x+y-2\log\left(2+x+y\right)=x+C$$

$$y-2\log\left(2+x+y\right)=C$$

$x=1 \qquad y=0$

$$-2\log 3=C$$

$$y-2\log\left(2+x+y\right)=-2\log 3$$

$$y=2\log\left(\frac{2+x+y}{3}\right)$$

at $x=6$

$$y=2\log\left(\frac{8+y}{3}\right)$$

$$\frac{8+y}{3}=e^{\frac{y}{2}}$$

$$8+y=3e^{\frac{y}{2}}$$

$$y<e^{\frac{y}{2}}$$

$$8+e^{\frac{y}{2}}\geq 3e^{\frac{y}{2}}$$

$$8\geq 2e^{\frac{y}{2}}$$

$$e^{\frac{y}{2}}\leq 4$$

$$\frac{y}{2}\leq \ell n4$$

$y \le 2\ell n4$

31. Match the following column :

Column I	**Column II**
(A) Area of region formed by points (x, y) satisfying $[x] = [y]$, $0 \le x, y \le 10$, [.] denote greatest integer function is	(P) 2
(B) Area of region formed by points (x, y) satisfying $\max(|x|, |y|) \le 1$ is	(Q) 4
(C) If area of region formed by points (x, y) satisfying $|y + x| \le 1$, $|y - x| \le 1$ and $2x^2 + 2y^2 \ge 1$ is $k - \frac{\pi}{2}$, then k =	(R) 6
(D) If area of region formed by points (x, y) satisfying $4 \le x^2 + y^2 \le 2(|x| + |y|)$ is 2k, then k =	(S) 10
	(T) 12

निम्न कॉलम को सुमेलित कीजिए :

Column I	**Column II**
(A) खा, ञ्ज्ञ ए0 $\le x, y \le 10$ को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल होगा	(P) 2
(B) उं ग ; ञ्ज्ञ ञ्ज्ञ ञ्ज्ञ ञ्ज्ञ ≤ 1 को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल होगा	(Q) 4
(C) ञ्ज्ञ गद्य ≤ 1, $|y - x| \le 1$ & $2x^2 + 2y^2 \ge 1$ को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल $k - \frac{\pi}{2}$, है तो k =	(R) 6
(D) $4 \le x^2 + y^2 \le 2(|x| + |y|)$ को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल 2k है तो k =	(S) 10
	(T) 12

Ans. A – S ; B – Q ; C – P ; D – Q

Sol. (A) Given : $[x] = [y]$

and $0 \le x, y \le 10$

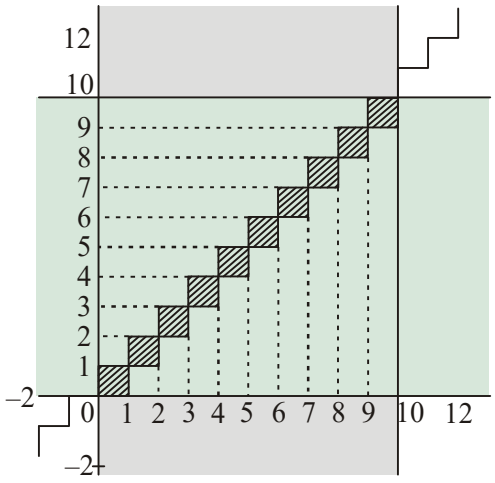
Case 1 $0 \le x < 1, 0 \le y < 1$

Case 2 $1 \le x < 2, 1 \le y < 2$

⋮

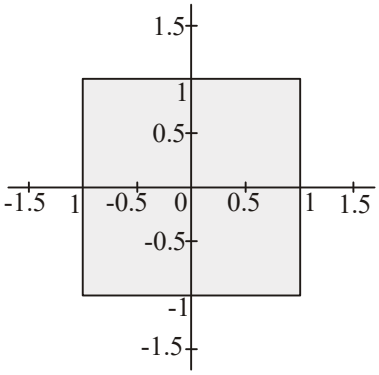
Case 10 $9 \le x < 10, 9 \le y < 10$

Similarly 10 cases wil for m; each case contain a square of area 1 sq. unit



Therefore required area is 10 sq. units.

(B) Given condition



$\max(|x|, |y|) \le 1$

There will be a square formed by lines

$x = \pm 1$ and $y = \pm 1$

So side of square = 2 units

required area = 4 sq. units

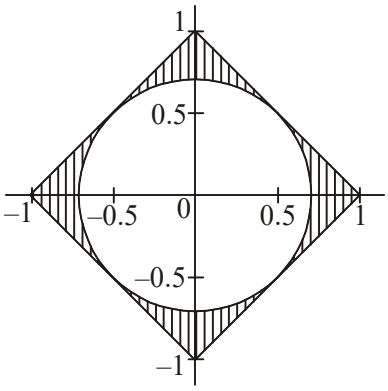
(C) Given constraints

$|y + x| \le 1$ _____(1)

$|y - x| \le 1$ _____(2)

fromo (1) and (2) a square will be formed having vertices $(0, \pm 1)$ and $(\pm 1, 0)$

So area of square $A_1 = 2$ sq. units



Also $x^2 + y^2 \ge \frac{1}{2}$

area of circle $A_2 = \pi \times \frac{1}{2}$

required area $= A_1 - A_2$

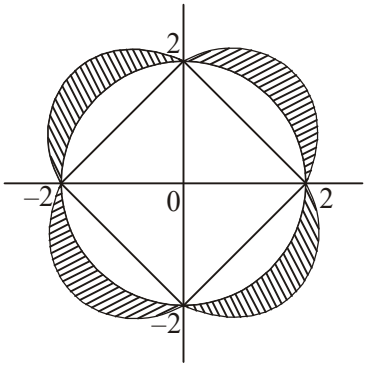
$= 2 - \frac{\pi}{2}$

$\therefore k = 2$

(D) Given constraints

$4 \le x^2 + y^2 \le 2(|x| + |y|)$

A_1 = area of square formed by $|x| + |y| \ge 2$



$A_1 = \left(2\sqrt{2}\right)^2 = 8$ square units

A_2 = area of circle formed by $x^2 + y^2 \ge 4$

$A_2 = \pi(4)$ square units

A_3 = area of shape for med by $x^2 + y^2 \le 2(|x| + |y|)$

$A_3 = 4 \times$ semi circles $= 4 \times \frac{\pi}{2} \times \left(\sqrt{2}\right)^2$

$= 4\pi$ square units

Required area $= A_3 - (A_2 - A_1)$

$= 4\pi - (4\pi - 8)$

$= 8$ square units

ATQ $2k = 8$

$k = 4$